

Contact

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www.linkedin.com/in/kevinkakolla
(LinkedIn)
github.com/torontodeveloper
(Portfolio)

Top Skills

LangChain

Retrieval-Augmented Generation
(RAG)

Generative AI

Languages

English (Full Professional)

Honors-Awards

Bantral Scholarship

Kevin Kakolla, MBA

GenAI / LLM Engineer | AI Agents, Voice Agent, Web Agents |
Python | CMU
Pittsburgh, Pennsylvania, United States

Summary

Senior AI / Machine Learning Engineer with 15+ years of experience building production AI and data systems across finance, telecom, healthcare, and energy.

I'm currently focused on GenAI, LLM applications, AI agents, RAG, and multimodal systems through graduate-level work at Carnegie Mellon University. My recent work includes building web/browser agents that convert natural-language instructions into actions on real websites, AI sales voice agent, fine tuning LLMs combining DOM candidate ranking, LLM action prediction, and end-to-end execution.

My background spans end-to-end ML pipelines, NLP, computer vision, conversational AI, and production software systems. I have hands-on experience with Python, PyTorch, TensorFlow, Scikit-learn, React, Django, and LLM-powered applications using models such as GPT-4o, Llama, and Mistral.

I've delivered AI and software solutions for organizations including Fleetworthy, RBC, TD Bank, Scotiabank, Ontario Power Generation, Bell Canada, Rogers, the Chan Zuckerberg Initiative, and Fujitsu — translating complex business problems into scalable, data-driven systems.

With an MBA and deep technical experience, I bridge technical execution with business strategy, helping teams move from prototypes to production-ready AI systems.

Open to Senior ML Engineer, AI Engineer, and LLM Engineer opportunities in Pittsburgh, open to relocation to the San Francisco Bay Area

Experience

Carnegie Mellon University

Graduate Researcher — Generative AI & LLMs | Carnegie Mellon University

January 2026 - Present (5 months)

Pittsburgh, Pennsylvania, United States

Enrolled in the Generative AI & LLM Graduate program at Carnegie Mellon University.

- Building a multimodal AI agent for language-guided web navigation using GPT-4o and Gemini 3.1
- Fine-tuning DeBERTa in candidate generation and GPT-4o/Gemini 3.1 on action prediction stage on the Mind2Web dataset to improve instruction-following and action grounding in web-based environments
- Working with multimodal and structured web interaction data to train agents capable of executing natural language instructions across complex UI environments

Fleetworthy

Machine Learning Engineer / Full Stack Developer

October 2025 - April 2026 (7 months)

Toronto, Ontario, Canada

Integrating Sales Voice Agent into Sales Lead Management for outbound calls with customers

Training Open source models like Llama 4.1 17B Maverick, Qwen locally on Nvidia Blackwell RTX 5090

Setting up local environment for Blackwell, Pytorch, CUDA 12.1 toolkit

Quantization of LLM using Unsloth, DPO(Direct Preference Optimization)

Inference And serving using vLLM, SGLang

Evaluate using BenchLocal, DeepEval, Ragas for

InstructFollow, ToolCall, StructOutput

- Built a React-based fleet management platform for truck/trailer inspection, user management, and site administration using Material UI, HTML, and CSS
- Implemented responsive design for iPad and tablet viewports
- Developed Django REST API endpoints and PostgreSQL schema for preclearance business logic
- Delivered full-stack features across Admin, Partner, Fleet, Vehicle, Inspection Site, and Fleet Manager modules
- Wrote unit tests using Jest, Vitest, and React Testing Library

T6 Health Systems

Machine Learning Engineer/Full Stack Developer

April 2025 - September 2025 (6 months)

Toronto, Ontario, Canada

- Built ML-powered health monitoring and medical decision support applications for radiation and environmental health workflows using Python, Scikit-learn, Pandas, Dask, MLflow, and Azure AI
- Developed scalable data pipelines on historical health, biomedical, and environmental sensor data to predict radiation exposure risk
- Integrated XGBoost, Ridge, and Lasso regression models for patient criticality and exposure-risk prediction with SHAP-based explainability
- Applied ResNet50 and VGG16 for biomedical image classification and environmental health monitoring
- Worked with whole slide imaging and HALO-based biomedical workflows, integrating image analysis with clinical decision-support systems
- Built React/TypeScript dashboards for real-time health monitoring across mobile and desktop platforms

Ontario Power Generation

Senior Machine Learning Engineer

June 2023 - April 2025 (1 year 11 months)

Greater Toronto Area, Canada

- Built ML recommendation models with confidence-score prediction for automated radiation permit (REP) assignment, reducing manual review overhead
- Developed internal and external ChatGPT-style applications with Basic, Advanced, and Guided conversation modes using GPT-3.5 Turbo, GPT-4, React, and Python FastAPI
- Implemented RAG-based intelligent search using Azure Cognitive Search, Azure OpenAI, and Pinecone VectorDB with prompt engineering for enterprise document retrieval
- Fine-tuned and deployed LLMs including Mistral 7B, Llama 2, Llama 3, Mistral 8x22B, and Stable Diffusion via Python FastAPI for text and image generation use cases
- Applied NLP classification using BERT, DistilBERT, RoBERTa, and DeBERTa on internal documents
- Trained and deployed ML models (classification, clustering, K-means) using Azure ML Studio, Vertex AI AutoML, and BigQuery ML
- Managed multi-source data pipelines across SQL Server, Oracle, and Azure Synapse Analytics for operational and analytical workloads

Scotiabank

Machine Learning/UI Developer

January 2023 - June 2023 (6 months)

Toronto, Ontario, Canada

- Built fraud detection models for credit card transactions using K-NN, Logistic Regression, and SVM with Scikit-learn
- Developed ChatBot application with Basic, Advanced, and Guided conversation modes using OpenAI GPT-3.5 Turbo, React, and Azure Databricks
- Implemented intelligent search using NLP techniques and LLMs to process and query banking documents
- Built ETL pipelines using Azure Databricks with cluster and pool configuration, and Apache Spark for data processing
- Fine-tuned OpenAI LLMs using zero-shot and few-shot learning for custom NLP applications
- Designed FastAPI microservices with authentication and monitoring for model serving
- Led a team of 5 front-end engineers delivering the GIC (Guaranteed Investment Certificate) Orion application in React, NextJS, and Styled Components
- Migrated legacy SOL system to Orion, redesigning the full GIC maturity flow and landing pages for Smart Investor and Mobile customers
- Automated CI/CD pipelines for model training and API deployment using Azure DevOps

Bell

Machine Learning Engineer

May 2021 - January 2023 (1 year 9 months)

Toronto, Ontario, Canada

Leading a team of 5 software engineers

Mentoring the team members

predict the churning of bell customers using supervised, unsupervised learning

Develop 360insights COOP/MDP application in ReactJS, Redux

- Features such as Login, Dashboard, menu navigation, MyFunding, My Marketing, Reports, Resources, Admn etc
- Building and usage of design system components 360ui which are used across all applications
- Using NextJS, ES6 javascript, HTML, CSS, react hooks, custom hooks
- Writing unit tests using react-testing-library, Jest

- Implementing Payment Solutions Framework to enable monetization – Checkout, Cart, Order Management, Order confirmation of printing solutions
 - Implementation of purchase and management of licenses/seats of HP Smart PRO subscription in React, Upgrade Trial etc
 - Using home made design system – veneer for css styling
 - Micro-front end implementation of Order Management, Cart Management, Checkout pages
- Developing UI screens for Bell technicians - Wifi checkup, Video services etc in ReactJS, javascript, Material UI, Redux

Goodfood Market (TSX:FOOD)

Data Scientist/ML Engineer

January 2021 - June 2021 (6 months)

Toronto, Ontario, Canada

Implemented search functionality to search products - recipes/groceries

Rewriting www.makegoodfood.ca using React, Javascript(Es6),reach-hook-forms, react-ui storybook, tailwind css, GraphQL technologies

Getting Images, Product categories (Groceries, Recipes), L1/L2 categories from Contentful and Algolia for indexing of products.

Writing unit tests using react-testing-library, Jest

Bell Media

Data Scientist/ML Engineer

September 2019 - December 2020 (1 year 4 months)

Toronto, Ontario, Canada

React developer to develop awesome User Interface using React, Javascript, HTML,CSS,Content Management System, Apollo GraphQL.

Re-wrote www.ctv.ca, www.crave.ca using React technologies and implement French Hub as well

Chan Zuckerberg Initiative(CZI)/Navigating Cancer

React Developer/ML Engineer

January 2019 - June 2019 (6 months)

Seattle, Washington, United States

Developing FORMS for TEAM MEETING, SCHEDULE CALL THRU HUBSPOT Form

Data is managed from CONTENTFUL Backend Server

Project is scaffolded using react-starter-kit

Data Analytics is configured and collected on Segment for Social Media platforms

Tickets Triage at hospitals for Doctors and Nurses

Created Visualization using horizontal box plot through D3 chart, React

Data collected for Scotia bank mortgage customer

Predictive Analysis using K-Nearest Neighbour algorithm

Toronto Dominion Bank

React/Angular Developer

August 2018 - January 2019 (6 months)

Toronto, Canada Area

Angular/React Developer for TD Mortgage Application

Equifax

Angular/React Developer

March 2018 - August 2018 (6 months)

Toronto, Canada Area

Angular UI Development for Equifax Interconnect Application

Currently using Angular 6, JavaScript, HTML5, CSS3, Bootstrap, Angular

Material, Typescript

Developing UI for Sasktel, BELL customers of Equifax

Scotiabank

Angular Developer

July 2017 - February 2018 (8 months)

Toronto, Canada Area

Angular based Trade Management back-office GUI

StreetSide,Block,Client Allocation, Back Office(Intact) screens development

REST API calls to backend, Angular 2, Angular 4, Bootstrap, Docker, HTML5,

Javascript

SCI MarketView

Angular UI Developer

January 2017 - July 2017 (7 months)

Markham

UI Development & Defect fixes of Sales Lead Application using Angular 1.x &

Angular 2.0, Angular UI Components, Bootstrap, HTML5, CSS3, JavaScript

RBC

Front End Developer

August 2016 - December 2016 (5 months)

Toronto, Canada Area

Java development,AngularJS, Spring

TD

Full Stack Developer(Tata Consultancy Services)

June 2016 - August 2016 (3 months)

Development, Business Analysis of Rogers Fido IPTV, High Speed Internet features

TD - SimpleApp Java development

Rogers Communications

IT Consultant (Tata Consultancy Services)

October 2015 - June 2016 (9 months)

Java, UI(AngualrJS) development for Rogers Fido Application

SAIT Polytechnic

Project Delivery Manager

June 2014 - June 2015 (1 year 1 month)

Managing the projects in Network, Storage, SAN technologies

System Analysis

Java/Network development

Haskayne School of Business

Full time MBA Student

September 2012 - December 2014 (2 years 4 months)

Subject courses Strategy, Corporate Finance, Accounting, HR, Project external issues, Supply chain procurement, Technology

Global Thermoelectric Inc.

Marketing Analyst

April 2014 - June 2014 (3 months)

Marketing analysis of Oil and Gas Pipeline,Upstream projects all over the world to determine the opportunities for ThermoGlobal remote power products

Polycom/TATA Elxsi

Technical Architect/Project Manager

January 2012 - March 2014 (2 years 3 months)

- Worked on a 3 month Contractual project in a team of 25 people

- Requirements Gathering, analyzing requirements through Use Cases, Traceability Matrix, Preparing Design Documents, Architecture for \$10 Million Project
- Develop impact and preliminary analyses, and participate actively in their implementation
- Participate in development projects as advisor in the area of business and systems use
- Negotiate with suppliers as part of Supply chain Procurement management
- Procuring material goods from suppliers through various Contract Pricing Strategy
- Used effectively Joint Application Development (JAD) to accelerate design process
- Observe the standards and procedures established by the project office
- Develop testing strategies and verification scenarios

Time Warner Cable/Union Pacific Railroad IT Consultant

April 2006 - December 2011 (5 years 9 months)

Time Warner Cable (contract)

Java/J2EE Designer

Led multiple global projects in the area of Mobile Technology iPhone, iPad, Blackberry and Android for client Time Warner Cable.

Design and development of Java/J2EE applications

Mentoring & Setting goals for the programmers

Dish Network

Java/J2EE Designer USA

Managed the Project of Middleware Services and R-connect vendors

Union Pacific/JP Morgan Chase

Java/J2EE Developer/Designer USA

Implementation of EPAY 2.0(Electronic Payment) into Enrolment Manager using WebServices (SOAP), JSP, JSTL, Java, stored procedures.

POJO's and EJB's implementation using IBM WebSphere

Re-factoring of Standard Client Profile (SCP) with J2EE design patterns, reorganization of code.

O-R mapping using iBatis and Hibernate

Architecture of Insight Project for Data warehouse

Generating Daily Sales Report for QuarterToDate for all the Seasons.

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Infosys

Project Manager

May 2004 - March 2005 (11 months)

Improving Equant GOLD (Global Order Delivery) system performance using Objected and Inventory Management, Order Management and estimates.

Currently Managing & Implementing Performance improvement measure for Order Management System implemented in Trilogy Weblogic Server.

Education

Carnegie Mellon University

Graduate Specialization — Generative AI & Large Language Models, Artificial Intelligence · (January 2026 - December 2026)

University of Toronto

Data Science/Machine Learning BootCamp, Data Science · (April 2025 - July 2025)

University of California, Berkeley

Program in Data Science, Artificial Intelligence · (October 2024 - December 2024)

University of California, Berkeley

Certificate, Large Language Model Agents · (September 2024 - December 2024)

University of Toronto

Artificial Intelligence Certificate, Artificial Intelligence · (September 2023 - September 2024)